THE KUNNETH SPECTRAL SEQUENCE FOR A BOUNDED COMPLEX OVER A RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Truncation of a complex and the identification of the associated graded pieces of the truncation filtration with the shifted cohomology. the author cites Homology, Section homology-section-truncations at proof lines 15812-15815. Supplies the filtered complex and the graded pieces that produce the Tor terms.
- The spectral sequence attached to a filtered complex with a two-stage filtration of the tensor product. the author cites Proposition proposition-kunneth-filtered at proof lines 15815-15818. Gives the spectral sequence whose first page is displayed.
- Reindexing of the bidegrees of a spectral sequence and the resulting shift of the differential. used at proof lines 15824-15832. Turns the first page of the constructed sequence into the Tor expression of the statement.

Source reference. Original lemma 0H7Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21474–21474. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0118: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1354–1354. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0H7Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21473–21473. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2 (Künneth Spectral Sequence)**.** *Let R be a ring. Let K and L be objects of $D^b(R)$. There exists a bigraded bounded spectral sequence $\{E_r\}_{r\geq 2}$ with*

$$E_2^{p,q} = \bigoplus\nolimits_{i+j=q} Tor^R_{-p}(H^i(K), H^j(L))$$

and d_r of bidegree $(r, -r+1)$ converging to $H^{p+q}(K \otimes^{\mathbf{L}}_R L)$.

Proof. Let $K^\bullet$ be a complex of R-modules representing K and let $L^\bullet$ be a complex of R-modules representing L. Set $F^iK^\bullet = \tau_{\leq -i}K^\bullet$ and similarly for L, see Homology, Section 0118. Apply Proposition 0H7Y noting that $\mathrm{gr}^iK^\bullet = H^{-i}(K)[i]$ to get a bounded spectral sequence $\{(E')_r\}_{r\geq 1}$ with

$$(E')_1^{p,q} = \bigoplus\nolimits_{i+j=p} \mathrm{Tor}^R_{-2p-q}(H^{-i}(K), H^{-j}(L))$$

converging to the cohomology of $K \otimes^{\mathbf{L}}_R L$. By construction this spectral sequence has differentials $d_r^{p,q} : (E')_r^{p,q} \to (E')_r^{p+r,q-r+1}$ for $r \geq 1$. To get the spectral sequence of the lemma we set for $r \geq 2$

$$E_r^{p,q} = (E')_{r-1}^{-q,p+2q}$$

We leave it to the reader to show that this works. □